

Chapter

15

Multiple Deposit Creation and the Money Supply Process

PREVIEW

As we saw in Chapter 5 and will see in later chapters on monetary theory, movements in the money supply affect interest rates and the overall health of the economy and thus affect us all. Because of its far-reaching effects on economic activity, it is important to understand how the money supply is determined. Who controls it? What causes it to change? How might control of it be improved? In this and subsequent chapters, we answer these questions by providing a detailed description of the *money supply process*, the mechanism that determines the level of the money supply.

Because deposits at banks are by far the largest component of the money supply, understanding how these deposits are created is the first step in understanding the money supply process. This chapter provides an overview of how the banking system creates deposits, and describes the basic principles of the money supply, needed to understand later chapters.

Four Players in the Money Supply Process

The “cast of characters” in the money supply story is as follows:

1. **The central bank**—the government agency that oversees the banking system and is responsible for the conduct of monetary policy; in the United States, it is called the Federal Reserve System
2. **Banks (depository institutions)**—the financial intermediaries that accept deposits from individuals and institutions and make loans: commercial banks, savings and loan associations, mutual savings banks, and credit unions
3. **Depositors**—individuals and institutions that hold deposits in banks
4. **Borrowers** from banks—individuals and institutions that borrow from the depository institutions and institutions that issue bonds that are purchased by the depository institutions

Of the four players, the central bank—the Federal Reserve System—is the most important. The Fed’s conduct of monetary policy involves actions that affect its balance sheet (holdings of assets and liabilities), to which we turn now.

The Fed's Balance Sheet

www.federalreserve.gov/boarddocs/rptcongress/annual01/default.htm

See the most recent Federal Reserve financial statement.

The operation of the Fed and its monetary policy involve actions that affect its balance sheet, its holdings of assets and liabilities. Here we discuss a **simplified balance sheet** that includes just four items that are essential to our understanding of the money supply process.¹

FEDERAL RESERVE SYSTEM	
Assets	Liabilities
Government securities	Currency in circulation
Discount loans	Reserves

Liabilities

www.rich.frb.org/research/econed/museum/

A virtual tour of the Federal Reserve's money museum.

The two liabilities on the balance sheet, currency in circulation and reserves, are often referred to as the **monetary liabilities of the Fed**. They are an important part of the money supply story, because increases in either or both will lead to an increase in the money supply (everything else being constant). The sum of the Fed's monetary liabilities (currency in circulation and reserves) and the U.S. Treasury's monetary liabilities (Treasury currency in circulation, primarily coins) is called the **monetary base**. When discussing the monetary base, we will focus only on the monetary liabilities of the Fed because the monetary liabilities of the Treasury account for less than 10% of the base.²

1. *Currency in circulation.* The Fed issues currency (those green-and-gray pieces of paper in your wallet that say "Federal Reserve Note" at the top). Currency in circulation is the amount of currency in the hands of the public. (Currency held by depository institutions is also a liability of the Fed, but is counted as part of the reserves.)

Federal Reserve notes are IOUs from the Fed to the bearer and are also liabilities, but unlike most, they promise to pay back the bearer solely with Federal Reserve notes; that is, they pay off IOUs with other IOUs. Accordingly, if you bring a \$100 bill to the Federal Reserve and demand payment, you will receive two \$50s, five \$20s, ten \$10s, or one hundred \$1 bills.

People are more willing to accept IOUs from the Fed than from you or me because Federal Reserve notes are a recognized medium of exchange; that is, they are accepted as a means of payment and so function as money. Unfortunately, neither you nor I can convince people that our IOUs are worth anything more than the paper they are written on.³

¹A detailed discussion of the Fed's balance sheet and the factors that affect the monetary base can be found in the appendix to this chapter, which you can find on this book's web site at www.aw.com/mishkin.

²It is also safe to ignore the Treasury's monetary liabilities when discussing the monetary base because the Treasury cannot actively supply its monetary liabilities to the economy due to legal restrictions.

³The currency item on our balance sheet refers only to currency *in circulation*; that is, the amount in the hands of the public. Currency that has been printed by the U.S. Bureau of Engraving and Printing is not automatically a liability of the Fed. For example, consider the importance of having \$1 million of your own IOUs printed up. You give out \$100 worth to other people and keep the other \$999,900 in your pocket. The \$999,900 of IOUs does not make you richer or poorer and does not affect your indebtedness. You care only about the \$100 of liabilities from the \$100 of circulated IOUs. The same reasoning applies for the Fed in regard to its Federal Reserve notes.

For similar reasons, the currency component of the money supply, no matter how it is defined, includes only currency in circulation. It does not include any additional currency that is not yet in the hands of the public. The fact that currency has been printed but is not circulating means that it is not anyone's asset or liability and thus cannot affect anyone's behavior. Therefore, it makes sense not to include it in the money supply.

2. *Reserves.* All banks have an account at the Fed in which they hold deposits. **Reserves** consist of deposits at the Fed plus currency that is physically held by banks (called *vault cash* because it is stored in bank vaults). Reserves are assets for the banks but liabilities for the Fed, because the banks can demand payment on them at any time and the Fed is required to satisfy its obligation by paying Federal Reserve notes. As you will see, an increase in reserves leads to an increase in the level of deposits and hence in the money supply.

Total reserves can be divided into two categories: reserves that the Fed requires banks to hold (**required reserves**) and any additional reserves the banks choose to hold (**excess reserves**). For example, the Fed might require that for every dollar of deposits at a depository institution, a certain fraction (say, 10 cents) must be held as reserves. This fraction (10%) is called the **required reserve ratio**. Currently, the Fed pays no interest on reserves.

Assets

The two assets on the Fed's balance sheet are important for two reasons. First, changes in the asset items lead to changes in reserves and consequently to changes in the money supply. Second, because these assets (government securities and discount loans) earn interest while the liabilities (currency in circulation and reserves) do not, the Fed makes billions of dollars every year—its assets earn income, and its liabilities cost nothing. Although it returns most of its earnings to the federal government, the Fed does spend some of it on “worthy causes,” such as supporting economic research.

1. *Government securities.* This category of assets covers the Fed's holdings of securities issued by the U.S. Treasury. As you will see, the Fed provides reserves to the banking system by purchasing securities, thereby increasing its holdings of these assets. An increase in government securities held by the Fed leads to an increase in the money supply.

2. *Discount loans.* The Fed can provide reserves to the banking system by making discount loans to banks. An increase in discount loans can also be the source of an increase in the money supply. The interest rate charged banks for these loans is called the **discount rate**.

Control of the Monetary Base

The *monetary base* (also called **high-powered money**) equals currency in circulation C plus the total reserves in banking system R .⁴ The monetary base MB can be expressed as

$$MB = C + R$$

The Federal Reserve exercises control over the monetary base through its purchases or sale of government securities in the open market, called **open market operations**, and through its extension of discount loans to banks.

Federal Reserve Open Market Operations

The primary way in which the Fed causes changes in the monetary base is through its open market operations. A purchase of bonds by the Fed is called an **open market purchase**, and a sale of bonds by the Fed is called an **open market sale**.

⁴Here currency in circulation includes both Federal Reserve currency (Federal Reserve notes) and Treasury currency (primarily coins).

Open Market Purchase from a Bank. Suppose that the Fed purchases \$100 of bonds from a bank and pays for them with a \$100 check. The bank will either deposit the check in its account with the Fed or cash it in for currency, which will be counted as vault cash. To understand what occurs as a result of this transaction, we look at *T-accounts*, which list only the changes that occur in balance sheet items starting from the initial balance sheet position. Either action means that the bank will find itself with \$100 more reserves and a reduction in its holdings of securities of \$100. The T-account for the banking system, then, is:

BANKING SYSTEM			
Assets		Liabilities	
Securities	−\$100		
Reserves	+\$100		

The Fed meanwhile finds that its liabilities have increased by the additional \$100 of reserves, while its assets have increased by the \$100 of additional securities that it now holds. Its T-account is:

FEDERAL RESERVE SYSTEM			
Assets		Liabilities	
Securities	+\$100	Reserves	+\$100

The net result of this open market purchase is that reserves have increased by \$100, the amount of the open market purchase. Because there has been no change of currency in circulation, the monetary base has also risen by \$100.

Open Market Purchase from the Nonbank Public. To understand what happens when there is an open market purchase from the nonbank public, we must look at two cases. First, let's assume that the person or corporation that sells the \$100 of bonds to the Fed deposits the Fed's check in the local bank. The nonbank public's T-account after this transaction is:

NONBANK PUBLIC			
Assets		Liabilities	
Securities	−\$100		
Checkable deposits	+\$100		

When the bank receives the check, it credits the depositor's account with the \$100 and then deposits the check in its account with the Fed, thereby adding to its reserves. The banking system's T-account becomes:

BANKING SYSTEM			
Assets		Liabilities	
Reserves	+\$100	Checkable deposits	+\$100

The effect on the Fed's balance sheet is that it has gained \$100 of securities in its assets column, while it has an increase of \$100 of reserves in its liabilities column:

FEDERAL RESERVE SYSTEM			
Assets		Liabilities	
Securities	+\$100	Reserves	+\$100

As you can see in the above T-account, when the Fed's check is deposited in a bank, the net result of the Fed's open market purchase from the nonbank public is identical to the effect of its open market purchase from a bank: Reserves increase by the amount of the open market purchase, and the monetary base increases by the same amount.

If, however, the person or corporation selling the bonds to the Fed cashes the Fed's check either at a local bank or at a Federal Reserve bank for currency, the effect on reserves is different.⁵ This seller will receive currency of \$100 while reducing holdings of securities by \$100. The bond seller's T-account will be:

NONBANK PUBLIC			
Assets		Liabilities	
Securities	-\$100		
Currency	+\$100		

The Fed now finds that it has exchanged \$100 of currency for \$100 of securities, so its T-account is:

FEDERAL RESERVE SYSTEM			
Assets		Liabilities	
Securities	+\$100	Currency in circulation	+\$100

⁵If the bond seller cashes the check at the local bank, its balance sheet will be unaffected, because the \$100 of vault cash that it pays out will be exactly matched by the deposit of the \$100 check at the Fed. Thus its reserves will remain the same, and there will be no effect on its T-account. That is why a T-account for the banking system does not appear here.

The net effect of the open market purchase in this case is that reserves are unchanged, while currency in circulation increases by the \$100 of the open market purchase. Thus the monetary base increases by the \$100 amount of the open market purchase, while reserves do not. This contrasts with the case in which the seller of the bonds deposits the Fed's check in a bank; in that case, reserves increase by \$100, and so does the monetary base.

The analysis reveals that *the effect of an open market purchase on reserves depends on whether the seller of the bonds keeps the proceeds from the sale in currency or in deposits*. If the proceeds are kept in currency, the open market purchase has no effect on reserves; if the proceeds are kept as deposits, reserves increase by the amount of the open market purchase.

The effect of an open market purchase on the monetary base, however, is always the same (the monetary base increases by the amount of the purchase) whether the seller of the bonds keeps the proceeds in deposits or in currency. The impact of an open market purchase on reserves is much more uncertain than its impact on the monetary base.

Open Market Sale. If the Fed sells \$100 of bonds to a bank or the nonbank public, the monetary base will decline by \$100. For example, if the Fed sells the bonds to an individual who pays for them with currency, the buyer exchanges \$100 of currency for \$100 of bonds, and the resulting T-account is:

NONBANK PUBLIC			
Assets		Liabilities	
Securities	+\$100		
Currency	−\$100		

The Fed, for its part, has reduced its holdings of securities by \$100 and has also lowered its monetary liability by accepting the currency as payment for its bonds, thereby reducing the amount of currency in circulation by \$100:

FEDERAL RESERVE SYSTEM			
Assets		Liabilities	
Securities	−\$100	Currency in circulation	−\$100

The effect of the open market sale of \$100 of bonds is to reduce the monetary base by an equal amount, although reserves remain unchanged. Manipulations of T-accounts in cases in which the buyer of the bonds is a bank or the buyer pays for the bonds with a check written on a checkable deposit account at a local bank lead to the same \$100 reduction in the monetary base, although the reduction occurs because the level of reserves has fallen by \$100.

Study Guide

The best way to learn how open market operations affect the monetary base is to use T-accounts. Using T-accounts, try to verify that an open market sale of \$100 of bonds to a bank or to a person who pays with a check written on a bank account leads to a \$100 reduction in the monetary base.

The following conclusion can now be drawn from our analysis of open market purchases and sales: ***The effect of open market operations on the monetary base is much more certain than the effect on reserves.*** Therefore, the Fed can control the monetary base with open market operations more effectively than it can control reserves.

Open market operations can also be done in other assets besides government bonds and have the same effects on the monetary base we have described here. One example of this is a foreign exchange intervention by the Fed (see Box 1).

Shifts from Deposits into Currency

Even if the Fed does not conduct open market operations, a shift from deposits to currency will affect the reserves in the banking system. However, such a shift will have no effect on the monetary base, another reason why the Fed has more control over the monetary base than over reserves.

Let's suppose that Jane Brown (who opened a \$100 checking account at the First National Bank in Chapter 9) decides that tellers are so abusive in all banks that she closes her account by withdrawing the \$100 balance in cash and vows never to deposit it in a bank again. The effect on the T-account of the nonbank public is:

NONBANK PUBLIC	
Assets	Liabilities
Checkable deposits	−\$100
Currency	+\$100

Box 1: Global**Foreign Exchange Rate Intervention and the Monetary Base**

It is common to read in the newspaper about a Federal Reserve intervention to buy or sell dollars in the foreign exchange market. (Note that this intervention occurs at the request of the U.S. Treasury.) Can this intervention also be a factor that affects the monetary base? The answer is yes, because a Federal Reserve intervention in the foreign exchange market involves a purchase or sale of assets denominated in a foreign currency.

Suppose that the Fed purchases \$100 of deposits denominated in euros in exchange for \$100 of deposits

at the Fed (a sale of dollars for euros). A Federal Reserve purchase of any asset, whether it is a U.S. government bond or a deposit denominated in a foreign currency, is still just an open market purchase and so leads to an equal rise in the monetary base. Similarly, a sale of foreign currency deposits is just an open market sale and leads to a decline in the monetary base. Federal Reserve interventions in the foreign exchange market are thus an important influence on the monetary base, a topic that we discuss further in Chapter 20.

The banking system loses \$100 of deposits and hence \$100 of reserves:

BANKING SYSTEM			
Assets		Liabilities	
Reserves	−\$100	Checkable deposits	−\$100

For the Fed, Jane Brown’s action means that there is \$100 of additional currency circulating in the hands of the public, while reserves in the banking system have fallen by \$100. The Fed’s T-account is:

FEDERAL RESERVE SYSTEM			
Assets		Liabilities	
		Currency in circulation	+\$100
		Reserves	−\$100

The net effect on the monetary liabilities of the Fed is a wash; the monetary base is unaffected by Jane Brown’s disgust at the banking system. But reserves are affected. Random fluctuations of reserves can occur as a result of random shifts into currency and out of deposits, and vice versa. The same is not true for the monetary base, making it a more stable variable.

Discount Loans

In this chapter so far we have seen changes in the monetary base solely as a result of open market operations. However, the monetary base is also affected when the Fed makes a discount loan to a bank. When the Fed makes a \$100 discount loan to the First National Bank, the bank is credited with \$100 of reserves from the proceeds of the loan. The effects on the balance sheets of the banking system and the Fed are illustrated by the following T-accounts:

BANKING SYSTEM				FEDERAL RESERVE SYSTEM			
Assets		Liabilities		Assets		Liabilities	
Reserves	+\$100	Discount loans	+\$100	Discount loans	+\$100	Reserves	+\$100

The monetary liabilities of the Fed have now increased by \$100, and the monetary base, too, has increased by this amount. However, if a bank pays off a loan from the Fed, thereby reducing its borrowings from the Fed by \$100, the T-accounts of the banking system and the Fed are as follows:

BANKING SYSTEM				FEDERAL RESERVE SYSTEM			
Assets		Liabilities		Assets		Liabilities	
Reserves	−\$100	Discount loans	−\$100	Discount loans	−\$100	Reserves	−\$100

The net effect on the monetary liabilities of the Fed, and hence on the monetary base, is then a reduction of \$100. We see that the monetary base changes one-for-one with the change in the borrowings from the Fed.

Other Factors That Affect the Monetary Base

So far in this chapter, it seems as though the Fed has complete control of the monetary base through its open market operations and discount loans. However, the world is a little bit more complicated for the Fed. Two important items that are not controlled by the Fed but affect the monetary base are *float* and *Treasury deposits at the Fed*. When the Fed clears checks for banks, it often credits the amount of the check to a bank that has deposited it (increases the bank’s reserves) but only later debits (decreases the reserves of) the bank on which the check is drawn. The resulting temporary net increase in the total amount of reserves in the banking system (and hence in the monetary base) occurring from the Fed’s check-clearing process is called **float**. When the U.S. Treasury moves deposits from commercial banks to its account at the Fed, leading to a rise in *Treasury deposits at the Fed*, it causes a deposit outflow at these banks like that shown in Chapter 9 and thus causes reserves in the banking system and the monetary base to fall. Thus *float* (affected by random events such as the weather, which affects how quickly checks are presented for payment) and *Treasury deposits at the Fed* (determined by the U.S. Treasury’s actions) both affect the monetary base but are not fully controlled by the Fed. Decisions by the U.S. Treasury to have the Fed intervene in the foreign exchange market also affect the monetary base, as can be seen in Box 1.

Overview of the Fed’s Ability to Control the Monetary Base

The factor that most affects the monetary base is the Fed’s holdings of securities, which are completely controlled by the Fed through its open market operations. Factors not controlled by the Fed (for example, float and Treasury deposits with the Fed) undergo substantial short-run variations and can be important sources of fluctuations in the monetary base over time periods as short as a week. However, these fluctuations are usually quite predictable and so can be offset through open market operations. *Although float and Treasury deposits with the Fed undergo substantial short-run fluctuations, which complicate control of the monetary base, they do not prevent the Fed from accurately controlling it.*

Multiple Deposit Creation: A Simple Model

With our understanding of how the Federal Reserve controls the monetary base and how banks operate (Chapter 9), we now have the tools necessary to explain how deposits are created. When the Fed supplies the banking system with \$1 of additional

reserves, deposits increase by a multiple of this amount—a process called **multiple deposit creation**.

**Deposit Creation:
The Single Bank**

Suppose that the \$100 open market purchase described earlier was conducted with the First National Bank. After the Fed has bought the \$100 bond from the First National Bank, the bank finds that it has an increase in reserves of \$100. To analyze what the bank will do with these additional reserves, assume that the bank does not want to hold excess reserves because it earns no interest on them. We begin the analysis with the following T-account:

FIRST NATIONAL BANK			
Assets		Liabilities	
Securities	−\$100		
Reserves	+\$100		

Because the bank has no increase in its checkable deposits, required reserves remain the same, and the bank finds that its additional \$100 of reserves means that its excess reserves have increased by \$100. Let's say that the bank decides to make a loan equal in amount to the \$100 increase in excess reserves. When the bank makes the loan, it sets up a checking account for the borrower and puts the proceeds of the loan into this account. In this way, the bank alters its balance sheet by increasing its liabilities with \$100 of checkable deposits and at the same time increasing its assets with the \$100 loan. The resulting T-account looks like this:

FIRST NATIONAL BANK			
Assets		Liabilities	
Securities	−\$100	Checkable deposits	+\$100
Reserves	+\$100		
Loans	+\$100		

The bank has created checkable deposits by its act of lending. Because checkable deposits are part of the money supply, the bank's act of lending has in fact created money. In its current balance sheet position, the First National Bank still has excess reserves and so might want to make additional loans. However, these reserves will not stay at the bank for very long. The borrower took out a loan not to leave \$100 idle at the First National Bank but to purchase goods and services from other individuals and corporations. When the borrower makes these purchases by writing checks, they will be deposited at other banks, and the \$100 of reserves will leave the First National Bank. *A bank cannot safely make loans for an amount greater than the excess reserves it has before it makes the loan.*

The final T-account of the First National Bank is:

FIRST NATIONAL BANK			
Assets		Liabilities	
Securities	− \$100		
Loans	+ \$100		

The increase in reserves of \$100 has been converted into additional loans of \$100 at the First National Bank, plus an additional \$100 of deposits that have made their way to other banks. (All the checks written on accounts at the First National Bank are deposited in banks rather than converted into cash, because we are assuming that the public does not want to hold any additional currency.) Now let's see what happens to these deposits at the other banks.

**Deposit Creation:
The Banking
System**

To simplify the analysis, let us assume that the \$100 of deposits created by First National Bank's loan is deposited at Bank A and that this bank and all other banks hold no excess reserves. Bank A's T-account becomes:

BANK A			
Assets		Liabilities	
Reserves	+ \$100	Checkable deposits	+ \$100

If the required reserve ratio is 10%, this bank will now find itself with a \$10 increase in required reserves, leaving it \$90 of excess reserves. Because Bank A (like the First National Bank) does not want to hold on to excess reserves, it will make loans for the entire amount. Its loans and checkable deposits will then increase by \$90, but when the borrower spends the \$90 of checkable deposits, they and the reserves at Bank A will fall back down by this same amount. The net result is that Bank A's T-account will look like this:

BANK A			
Assets		Liabilities	
Reserves	+ \$10	Checkable deposits	+ \$100
Loans	+ \$90		

If the money spent by the borrower to whom Bank A lent the \$90 is deposited in another bank, such as Bank B, the T-account for Bank B will be:

BANK B			
Assets		Liabilities	
Reserves	+\$90	Checkable deposits	+\$90

The checkable deposits in the banking system have increased by another \$90, for a total increase of \$190 (\$100 at Bank A plus \$90 at Bank B). In fact, the distinction between Bank A and Bank B is not necessary to obtain the same result on the overall expansion of deposits. If the borrower from Bank A writes checks to someone who deposits them at Bank A, the same change in deposits would occur. The T-accounts for Bank B would just apply to Bank A, and its checkable deposits would increase by the total amount of \$190.

Bank B will want to modify its balance sheet further. It must keep 10% of \$90 (\$9) as required reserves and has 90% of \$90 (\$81) in excess reserves and so can make loans of this amount. Bank B will make an \$81 loan to a borrower, who spends the proceeds from the loan. Bank B's T-account will be:

BANK B			
Assets		Liabilities	
Reserves	+\$ 9	Checkable deposits	+\$90
Loans	+\$81		

The \$81 spent by the borrower from Bank B will be deposited in another bank (Bank C). Consequently, from the initial \$100 increase of reserves in the banking system, the total increase of checkable deposits in the system so far is \$271 (= \$100 + \$90 + \$81).

Following the same reasoning, if all banks make loans for the full amount of their excess reserves, further increments in checkable deposits will continue (at Banks C, D, E, and so on), as depicted in Table 1. Therefore, the total increase in deposits from the initial \$100 increase in reserves will be \$1,000: The increase is tenfold, the reciprocal of the 0.10 reserve requirement.

If the banks choose to invest their excess reserves in securities, the result is the same. If Bank A had taken its excess reserves and purchased securities instead of making loans, its T-account would have looked like this:

BANK A			
Assets		Liabilities	
Reserves	+\$10	Checkable deposits	+\$100
Securities	+\$90		

Table 1 Creation of Deposits (assuming 10% reserve requirement and a \$100 increase in reserves)

Bank	Increase in Deposits (\$)	Increase in Loans (\$)	Increase in Reserves (\$)
First National	0.00	100.00	0.00
A	100.00	90.00	10.00
B	90.00	81.00	9.00
C	81.00	72.90	8.10
D	72.90	65.61	7.29
E	65.61	59.05	6.56
F	59.05	53.14	5.91
.	.	.	.
.	.	.	.
.	.	.	.
Total for all banks	1,000.00	1,000.00	100.00

When the bank buys \$90 of securities, it writes a \$90 check to the seller of the securities, who in turn deposits the \$90 at a bank such as Bank B. Bank B's checkable deposits rise by \$90, and the deposit expansion process is the same as before. ***Whether a bank chooses to use its excess reserves to make loans or to purchase securities, the effect on deposit expansion is the same.***

You can now see the difference in deposit creation for the single bank versus the banking system as a whole. Because a single bank can create deposits equal only to the amount of its excess reserves, it cannot by itself generate multiple deposit expansion. A single bank cannot make loans greater in amount than its excess reserves, because the bank will lose these reserves as the deposits created by the loan find their way to other banks. However, the banking system as a whole can generate a multiple expansion of deposits, because when a bank loses its excess reserves, these reserves do not leave the banking system even though they are lost to the individual bank. So as each bank makes a loan and creates deposits, the reserves find their way to another bank, which uses them to make additional loans and create additional deposits. As you have seen, this process continues until the initial increase in reserves results in a multiple increase in deposits.

The multiple increase in deposits generated from an increase in the banking system's reserves is called the **simple deposit multiplier**.⁶ In our example with a 10% required reserve ratio, the simple deposit multiplier is 10. More generally, the simple deposit multiplier equals the reciprocal of the required reserve ratio, expressed as a fraction ($10 = 1/0.10$), so the formula for the multiple expansion of deposits can be written as:

$$\Delta D = \frac{1}{r} \times \Delta R \quad (1)$$

⁶This multiplier should not be confused with the Keynesian multiplier, which is derived through a similar step-by-step analysis. That multiplier relates an increase in income to an increase in investment, whereas the simple deposit multiplier relates an increase in deposits to an increase in reserves.

Deriving the Formula for Multiple Deposit Creation

where ΔD = change in total checkable deposits in the banking system

r = required reserve ratio (0.10 in the example)

ΔR = change in reserves for the banking system (\$100 in the example)⁷

The formula for the multiple creation of deposits can also be derived directly using algebra. We obtain the same answer for the relationship between a change in deposits and a change in reserves, but more quickly.

Our assumption that banks do not hold on to any excess reserves means that the total amount of required reserves for the banking system RR will equal the total reserves in the banking system R :

$$RR = R$$

The total amount of required reserves equals the required reserve ratio r times the total amount of checkable deposits D :

$$RR = r \times D$$

Substituting $r \times D$ for RR in the first equation:

$$r \times D = R$$

and dividing both sides of the preceding equation by r gives us:

$$D = \frac{1}{r} \times R$$

Taking the change in both sides of this equation and using delta to indicate a change:

$$\Delta D = \frac{1}{r} \times \Delta R$$

which is the same formula for deposit creation found in Equation 1.

This derivation provides us with another way of looking at the multiple creation of deposits, because it forces us to look directly at the banking system as a whole rather than one bank at a time. For the banking system as a whole, deposit creation (or contraction) will stop only when all excess reserves in the banking system are gone; that is, the banking system will be in equilibrium when the total amount of required reserves equals the total amount of reserves, as seen in the equation $RR = R$. When $r \times D$ is substituted for RR , the resulting equation $R = r \times D$ tells us how high checkable deposits will have to be in order for required reserves to equal total reserves. Accordingly, a given level of reserves in the banking system determines the level of checkable deposits when the banking system is in equilibrium (when $ER = 0$); put another way, the given level of reserves supports a given level of checkable deposits.

⁷A formal derivation of this formula follows. Using the reasoning in the text, the change in checkable deposits is \$100 ($= \Delta R \times 1$) plus \$90 ($= \Delta R \times (1 - r)$) plus \$81 ($= \Delta R \times (1 - r)^2$) and so on, which can be rewritten as:

$$\Delta D = \Delta R \times [1 + (1 - r) + (1 - r)^2 + (1 - r)^3 + \dots]$$

Using the formula for the sum of an infinite series found in footnote 5 in Chapter 4, this can be rewritten as:

$$\Delta D = \Delta R \times \frac{1}{1 - (1 - r)} = \frac{1}{r} \times \Delta R$$

In our example, the required reserve ratio is 10%. If reserves increase by \$100, checkable deposits must rise to \$1,000 in order for total required reserves also to increase by \$100. If the increase in checkable deposits is less than this, say \$900, then the increase in required reserves of \$90 remains below the \$100 increase in reserves, so there are still excess reserves somewhere in the banking system. The banks with the excess reserves will now make additional loans, creating new deposits, and this process will continue until all reserves in the system are used up. This occurs when checkable deposits have risen to \$1,000.

We can also see this by looking at the T-account of the banking system as a whole (including the First National Bank) that results from this process:

BANKING SYSTEM			
Assets		Liabilities	
Securities	−\$ 100	Checkable deposits	+\$1,000
Reserves	+\$ 100		
Loans	+\$1,000		

The procedure of eliminating excess reserves by loaning them out means that the banking system (First National Bank and Banks A, B, C, D, and so on) continues to make loans up to the \$1,000 amount until deposits have reached the \$1,000 level. In this way, \$100 of reserves supports \$1,000 (ten times the quantity) of deposits.

Critique of the Simple Model

Our model of multiple deposit creation seems to indicate that the Federal Reserve is able to exercise complete control over the level of checkable deposits by setting the required reserve ratio and the level of reserves. The actual creation of deposits is much less mechanical than the simple model indicates. If proceeds from Bank A's \$90 loan are not deposited but are kept in cash, nothing is deposited in Bank B, and the deposit creation process stops dead in its tracks. The total increase in checkable deposits is only \$100—considerably less than the \$1,000 we calculated. So if some proceeds from loans are used to raise the holdings of currency, checkable deposits will not increase by as much as our streamlined model of multiple deposit creation tells us.

Another situation ignored in our model is one in which banks do not make loans or buy securities in the full amount of their excess reserves. If Bank A decides to hold on to all \$90 of its excess reserves, no deposits would be made in Bank B, and this would also stop the deposit creation process. The total increase in deposits would again be only \$100 and not the \$1,000 increase in our example. Hence if banks choose to hold all or some of their excess reserves, the full expansion of deposits predicted by the simple model of multiple deposit creation does not occur.

Our examples rightly indicate that the Fed is not the only player whose behavior influences the level of deposits and therefore the money supply. Banks' decisions regarding the amount of excess reserves they wish to hold, depositors' decisions regarding how much currency to hold, and borrowers' decisions on how much to borrow from banks can cause the money supply to change. In the next chapter, we stress the behavior and interactions of the four players in constructing a more realistic model of the money supply process.